SEASON 5 OVER THE WALL
PUSH THE POSSIBILITIES

ROUND 2 · HAIR CLEANSING · BUSINESS PROPOSAL

Parachute Naturale

WaterSmart

The shampoo that checks your water first

দোষ কি চুলের, নাকি পানির?

Is it your hair, or your water?



Team The Last BrainCell

Over The Wall Season 5 · Marico Bangladesh · October 2026

WaterSmart turns Bangladesh's hard, iron-rich water into Naturale's next growth platform

THE PROBLEM

83%

of 3,530 sampled wells exceeded iron or hardness screening levels (BGS/DPHE)

Minerals block lather and leave hair rough and dull. People who move city or water source notice it first.

THE IDEA

3-in-1

chelating formula + PaniCheck cap + Pani Map

A Naturale shampoo with a hardness test strip in every 340 ml cap and a QR code to your area's water profile.

WHY MARICO

#2

Naturale is the 2nd most penetrated shampoo brand in Bangladesh

Made on existing Mouchak lines (+45% shampoo batch capacity in FY26) and sold through Naturale's distribution.

THE BUSINESS

Tk 41 Cr

Year-3 net revenue; EBITDA-positive from Year 2

NPV +Tk 3.1 Cr and IRR 25% against a 15% hurdle. Peak funding Tk 9.8 Cr.

OUR ASK Fund R&D and a 6-month, 1,500-outlet pilot: Tk 3.5 Cr committed before the national go/no-go.

Sources: BGS & DPHE (2001) national hydrochemical survey, team analysis; Marico Bangladesh Annual Report FY2025-26; team financial model v4 (planning assumptions, see Appendix A).

Every option fights for hair; none is built for the water

What someone with “bad-water hair” can choose today

Mass shampoos
Sunsilk, Clinic Plus, Naturale, Pantene · Tk 83–103 per 100 ml
Benefit-led, not built for minerals

Premium and anti-dandruff
Dove, Clear, Head & Shoulders, L’Oréal · Tk 142–180 per 100 ml
Costs more; same water problem

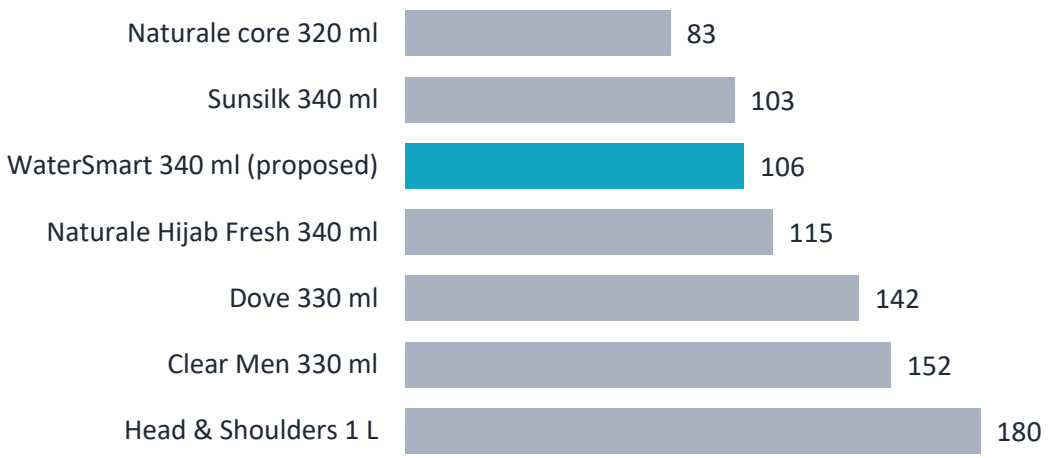
Imported hard-water shampoos
Dove Hard Water Defense (Gulf), Malibu C, import listings
Proves the need; import-priced, not in general trade

Workarounds
More shampoo, hair oil, conditioner, bottled-water rinse, filters
Cost more; treat symptoms only

Do nothing
Live with rough, dull hair
The biggest competitor

Tk ~2,150 Cr
Bangladesh shampoo market, 2026 estimate (USD 176M, Mordor Intelligence; Tk 122/USD)

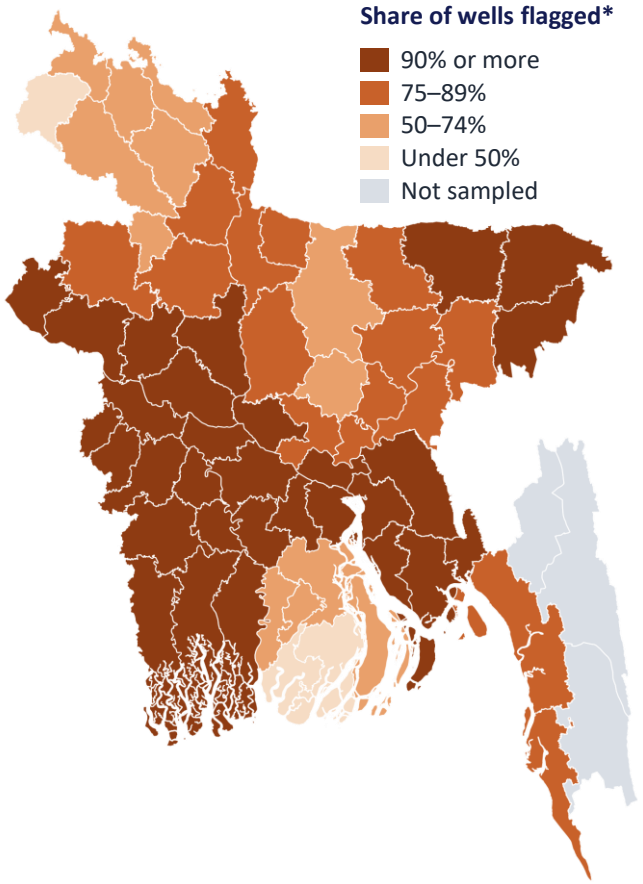
Shelf price per 100 ml (Tk), 9 Oct 2026



Unilever already sells “Hard Water Defense” in the Gulf, so the need is proven abroad. We found no mass-priced option on the Bangladeshi e-retailers we checked.

Sources: Chaldal and Arogga listings (accessed 9 Oct 2026, sale prices); Dove Arabia product page; Malibu C official site; Mordor Intelligence, Bangladesh Shampoo Market (2026 estimate).

The opportunity is in the water: in 58 of 61 sampled districts, most wells are iron-rich or hard



65% of 3,530 wells had iron above 0.3 mg/L, the level that stains buckets and tastes metallic

58% were hard or very hard: above 120 mg/L as $CaCO_3$ (USGS classes)

6 of 12 Dhaka tap locations tested hard in a 2025 study, up to 300 mg/L at Amin Bazar

5 of 5 largest division districts are majority-flagged: Dhaka 76%, Chattogram 86%, Rajshahi 94%, Khulna 96%, Sylhet 91%

Calcium, magnesium and iron deposit on hair and tie up shampoo surfactant → Less lather, a rough or dull feel, and more shampoo used per wash → **A chelator (GLDA) locks the minerals so they rinse away**

OUR TEST [date]: hall tap water [ ] mg/L hardness · [ ] mg/L iron · TDS [ ] ppm

**Iron >0.3 mg/L or hardness >120 mg/L. BGS & DPHE (2001): 3,534 wells sampled 1998–99, team analysis (historical wells, not today's taps). ICCEI 2025 Dhaka tap study; USGS hardness classes; US EPA iron guideline 0.3 mg/L.*

Movers feel it first: a new city, new water, and hair that no longer feels right

TARGET: ADULT "MOVERS", 18–35

- University and hall students: 4.82M in public universities and affiliated colleges (UGC, 2023)
- Mess residents and young workers who moved city or changed water source
- Urban and peri-urban, women and men; price-aware but brand-aware
- Where: hall, mess and rented-flat bathrooms on tube-well or supply water
- Want: hair that feels like home, proof rather than promises, a fair price
- Phase 2: coastal households on saline water (LonaJol)

#1 = function

Women shampoo users (n=390, Dhaka and Chattogram) rank functionality first (4.21/5), above brand image (4.15) and price (4.13). Packaging comes last (3.40).
Khanam & Rahman, 2022

47%

of coastal adults (182 of 391, Ramgati) say saline water frequently or very frequently causes hair loss, as they perceive it.
Zakaria et al., 2025

53.8M

Bangladeshis live in urban areas (31.7%). The cities Movers move to are majority-flagged for iron or hardness.

BBS Census 2022; BGS/DPHE

OUR CONSUMER SURVEY · 3–9 OCT 2026

- 49 consenting adults from 50 Google Form submissions
- 18 of 22 water-source changers reported worse hair problems (81.8%)
- 38 of 49 rated the concept 4 or 5 out of 5 (77.6%)
- Median maximum price for 340 ml: Tk 320 (n=47)
- 15 of 47 stated a maximum price of at least Tk 360 (31.9%)
- Convenience sample; self-reports. Water/foam tests remain planned.

INSIGHT People blame their hair, or their shampoo. The real culprit may be the water, and nobody gives them a way to check.

Sources: Khanam & Rahman (2022); Zakaria et al. (2025); UGC Annual Report 2023 via Prothom Alo (4 Oct 2025); BBS Census 2022. Team Google Form survey, 3–9 Oct 2026 (n=49; maximum-price answers n=47).

Parachute Naturale WaterSmart: check your water, then wash smarter



THE FORMULA · chelating cleanse

GLDA, a readily biodegradable chelator, binds iron, calcium and magnesium so they rinse away. Coconut-derived surfactants lather in hard water; Coconut Milk Protein conditions.

THE PACK · PaniCheck cap

A foil-sealed hardness strip in every 340 ml cap. Dip for one second, match the colour, and know your water in under a minute.

THE SERVICE · Pani Map

A QR code on every pack, sachets included, opens your area's water profile (BGS/DPHE data) and a step-by-step guide to testing your own water.

Sachet Tk 3 · 180 ml Tk 180 · 340 ml + PaniCheck Tk 360 · Phase 2: LonaJol (saline)

দোষ কি চুলের, নাকি পানির? আগে পানি চেক করুন।

Is it your hair, or your water? Check your water first.

Mock-ups are illustrative concepts. Product claims are proposed and subject to Marico R&D substantiation. GLDA: Nouryon Dissolvine product guide (readily biodegradable).

A Naturale variant, not a new brand: natural care upgraded with water science

BRAND PROMISE

Clean you can feel, in any water

Water science

GLDA chelation; lathers in hard water

Natural care

Coconut Milk Protein; paraben-free and dermatologically tested, like all Naturale

Know your water

PaniCheck strip and Pani Map

FOUNDATION Parachute Naturale, the 2nd most penetrated shampoo brand in Bangladesh

Personality

- Smart
- Honest
- Reassuring
- Youthful

CATEGORY FIT

Existing category: hair cleansing. New need-state: water-smart cleansing, the first Naturale variant chosen by your water rather than your hair type. It answers the brief's call for "hair cleansing solutions for emerging consumer needs and usage occasions".

WHY UNDER NATURALE

- 2nd most penetrated shampoo brand (Marico FY26)
- Same hero ingredient: Coconut Milk Protein
- Existing general trade, modern trade and e-commerce reach; e-commerce shampoo grew about 45% in FY26
- Cheaper to launch than a new brand

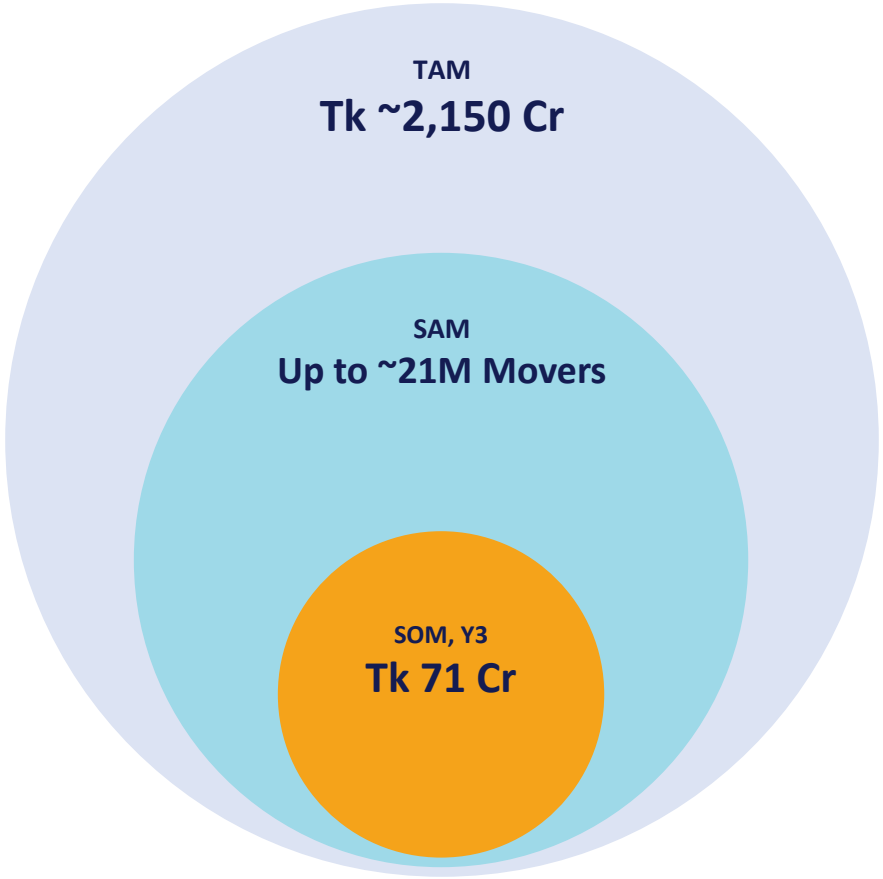
INNOVATION
Formula + a test in the cap

DIVERSIFICATION
New need-state; coastal Phase 2

SCALING UP
Existing lines; sachet-to-bottle ladder

Sources: Marico Bangladesh Annual Report FY2025-26 (Naturale penetration, e-commerce shampoo growth); OTW Season 5 case brief. Innovation, Diversification and Scaling Up are the Season 5 pivots.

A mass-market problem: a Tk ~2,150 Cr category and Tk 71 Cr of WaterSmart sales by Year 3



TAM

Tk ~2,150 Cr: Bangladesh shampoo market value, 2026 estimate (USD 176M, Mordor; Tk 122/USD). The problem is national: 58 of 61 sampled districts are majority-flagged for iron or hardness.

SAM

Up to ~21M Movers: 4.82M students in public universities and affiliated colleges (UGC 2023) plus ~16.5M lifetime internal migrants (9.7% of 169.8M people, 2011 census share). The groups overlap.

SOM

Tk 70.8 Cr gross sales in Year 3, 3.3% of today's category. 131M washes a year is about 1.3M regular users at two washes a week, roughly 6% of Movers.

Cross-checks: Year-3 net revenue Tk 41.4 Cr is 2.0% of Marico Bangladesh's FY26 revenue (Tk 2,071 Cr). Mass, not niche: a Tk 3 sachet entry and 104k outlets by Year 3.

Sources: Mordor Intelligence (2026e); UGC Annual Report 2023 via Prothom Alo; BBS Census 2011 (lifetime migration) and 2022 (population); Marico FY26; team model v4. Usage assumption: 7 ml per wash, two washes a week.

Our promise: hair that feels clean again, whatever water you live with

The Mover

JOB TO BE DONE

Get clean, soft hair with whatever water the hall, mess or flat has

PAINS

Low lather, a rough or sticky feel, dull hair, extra shampoo, orange stains, and “is it me or the water?”

GAINS

Hair that feels like home, confidence, no wasted money

FIT

WaterSmart

PRODUCTS

Iron & Hard Water shampoo, PaniCheck cap, Pani Map

PAIN RELIEVERS

Chelator removes mineral build-up*; lathers in hard water*; the strip ends the guesswork

GAIN CREATORS

Coconut Milk Protein care; Tk 106 per 100 ml against Tk 142–180 for premium brands; a Tk 3 sachet to try

For Movers whose hair changed with their water, WaterSmart is the Naturale shampoo that checks your water and cleans away iron and hard-water build-up, so hair feels like home again.

**Proposed claims, to be substantiated by Marico R&D (hair-tress deposit and lather tests) before launch. Prices: Chaldal and Arogga listings, 9 Oct 2026.*

USP: a water test in every cap. No shampoo we found offers one

1 Flip the cap
The strip sits in a foil pouch in the lid

2 Dip for 1 second
In your bucket, shower or tap water

3 Match the colour
Soft, moderate, hard or very hard

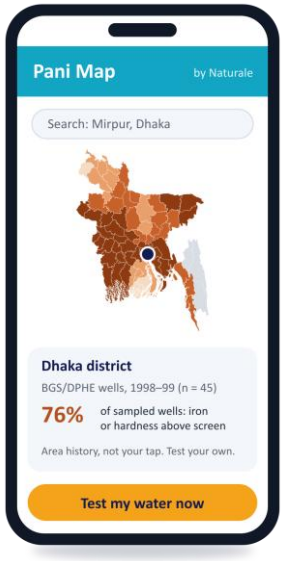
4 Scan the QR
Pani Map and tips for your water

Total hardness, mg/L as $CaCO_3$

0	60	120	250	425
Soft	Moderate	Hard	Very hard	

Illustrative colours; the final chart comes from the strip supplier.

Strip spec: total hardness 0–425 mg/L · individually foil-sealed · 18-month target life · Tk 3 target cost including sleeve



Feature	WaterSmart	Dove HWD (Gulf)	Naturale core	Mass brands
Mineral chelator	GLDA	EDTA	–	–
Water test in pack	✓	X	X	X
Area water info (QR)	✓	X	X	X
In Bangladesh GT	✓	X	✓	✓
Sachet trial	Tk 3	X	Tk 2	✓
Price per 100 ml	Tk 106	Import	Tk 83	Tk 103

See the problem, then solve it, in one pack.
Dove's Gulf pack shields hair from minerals. WaterSmart also shows people their problem, which turns every bottle into its own product demo.

Sources: Dove Arabia product page (ingredients include Disodium EDTA; no in-pack test shown); Chaldal and Arogga, 9 Oct 2026; Hach, Bartovation and Macherey-Nagel sealed hardness-strip specifications.

Campaign “Check your water first”: make the problem visible, then make WaterSmart the answer

BRAND TASK Build salience for a new need-state, “water hair”, among Movers, then turn curiosity into trial.

AWARE — Surround

#PaniCheckChallenge

Creators test hall and home water on camera on TikTok, Facebook and YouTube. Influencer-led, like Parachute Advansed Secrets.

CONSIDER — Unique

Pani Map on WhatsApp

Scan or message your upazila to see its water history and get a guide to testing your own water.

TRY — Unique

Campus “Pani Lab”

Booths at 30 universities and halls: a free strip test, a same-shampoo foam demo and Tk 3 sachet sampling.

REPEAT — Surround

Every bottle is a demo

The in-cap test works at home. E-commerce brand days and bundles; shelf strips with QR codes at cosmetics stores and supermarkets.

Launch calendar, Year 1

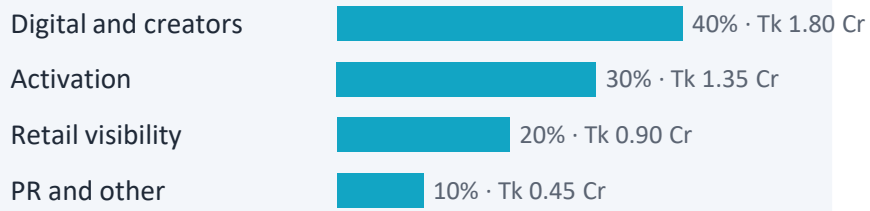
Q1
Pipeline fill and teaser

Q2
Campus blitz at session start

Q3
Creators and e-commerce brand days

Q4
Rural sachet push and haat activations

Year-1 A&P: Tk 4.5 Cr



KPIs: spontaneous awareness among Movers · consideration · water tests done (one per 340 ml bottle: 1.5M in three years) · trial · rate of sale. Framework: Marketing masterclass (unique and surround touchpoints). Budget split is a planning assumption.

Sales: sachets for reach, 340 ml for value: 37k outlets in Year 1, 104k by Year 3

Channel	Year-1 outlets	Year-3 outlets	Lead pack	Units / outlet / month (sachet · 180 · 340)
Rural grocer (RG)	15,000	45,000	Sachet	48 · 0.5 · –
Urban neighbourhood grocer (UNG)	12,000	35,000	Sachet, 180 ml	36 · 1 · 0.5
Urban cosmetics store (UCS)	6,000	12,000	340 ml + strip	– · 2 · 2
Rural cosmetics store (RCS)	3,000	9,000	180 ml	12 · 1 · 1
Wholesale (WS)	1,000	3,000	Sachet	240 · 3 · 1
Modern trade (MT)	300	360	340 ml	– · 15 · 15
E-commerce	30	40	340 ml	– · 30 · 40
Total	**37,330**	**104,400**		

Year-3 net revenue by channel



General trade Tk 38.5 Cr (93%) · Modern trade Tk 2.3 Cr · E-commerce Tk 0.6 Cr

BENCHMARKS (3 WAYS)

Existing product: Naturale Anti Hair Fall 320 ml, Tk 265

Similar channel: the Naturale range in GT, MT and e-commerce

Similar price point: Naturale Hijab Fresh 340 ml Tk 390; Sunsilk 340 ml Tk 350

BOOSTERS

TLO: 12-sachet display strips at launch

QPDS: shelf talker and QR at UCS and MT

TO/CO: e-commerce bundles and brand days

Incentive: “Dhamaka” bonus on 340 ml drops

Margins: retailer + distributor 22% on bottles, 25% on sachets

Outlet universe and channel types: Marico Sales masterclass (294k outlets in the demonstration). Units per outlet are planning assumptions to be validated in the pilot. Prices: Chaldal, 9 Oct 2026.

Supply chain: Make and Modify on existing Naturale lines; Buy only the strip and the chelator

PLAN

Demand set from pilot rate of sale; make-to-stock with 30 days' safety stock of strips and caps

SOURCE

Local: surfactants, bottles, caps, labels. Imported: GLDA (via local agents of Nouryon or BASF) and foil-sealed strips from two suppliers

MAKE

Mouchak: batch mixing (+45% capacity in FY26), bottle and sachet filling (two new sachet machines), strip insertion (manual in pilot, semi-automatic nationally)

MOVE

Existing Naturale distributors to GT, MT and e-commerce. QC release: pH 5.0–5.5, viscosity, microbiology, strip CoA and seal check

MAKE · MODIFY · BUY

MAKE Bulk shampoo; bottle and sachet filling

MODIFY Naturale base + GLDA; 340 ml flip-top cap with a strip pocket (new mould)

BUY Hardness strips with foil sleeve; GLDA chelator

FACTORY READINESS

- ✓ **Feasibility** Shampoo variant on an existing line
- ✓ **Affordability** Blended gross margin 47%
- ✓ **Capability** Make + Modify in-house
- ! **Time** 4–6 months of stability testing
- ✓ **Operation** 0.86M litres in Year 3
- ! **Risk** Strip life in humidity; DGDA and BSTI
- ✓ **Yield & scale** One platform, two variants

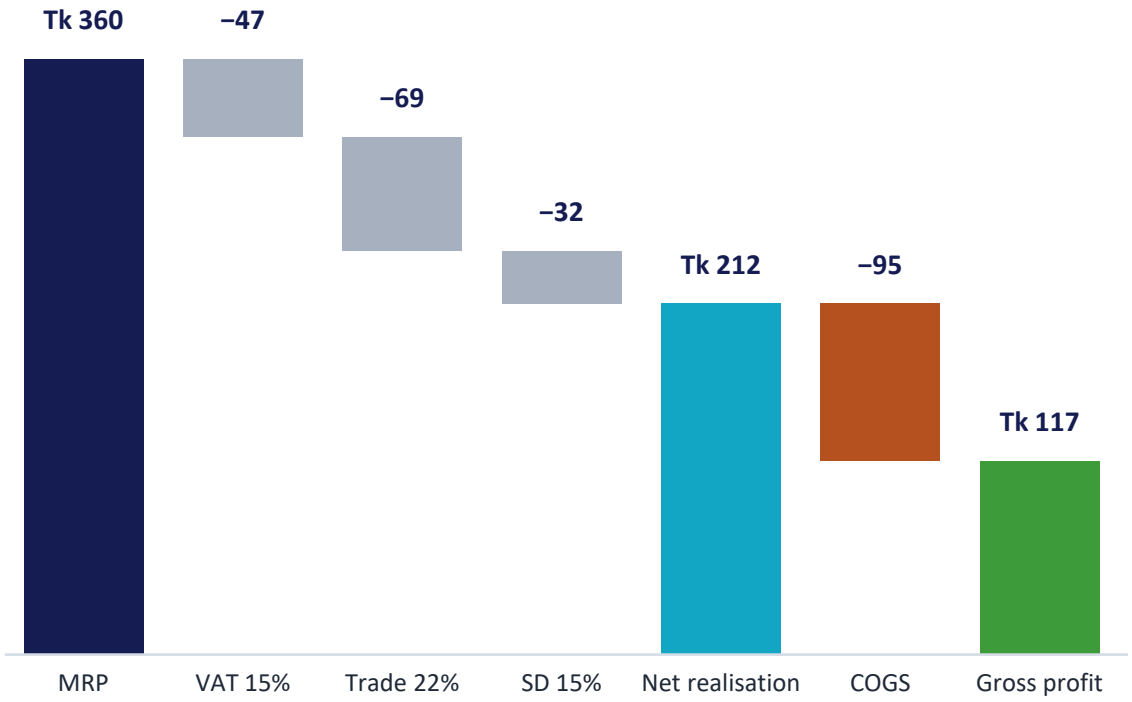
REGULATION & SUSTAINABILITY

- BSTI: shampoo needs certification under BDS 1269:2022
- DGDA: Drugs and Cosmetics Act 2023; 2026 cosmetics rules still in draft
- EPR Guidelines 2026: plastic take-back; 1% of net revenue reserved
- GLDA is readily biodegradable, unlike EDTA
- Phase 2 refill pouch: 68% less plastic per wash

Sources: Marico Bangladesh Annual Report FY2025-26 (Mouchak capacity); BSTI mandatory-certification list; Drugs and Cosmetics Act 2023; EPR Guidelines 2026 (SRO 293); Nouryon Dissolvine guide; Supply chain and Manufacturing masterclasses.

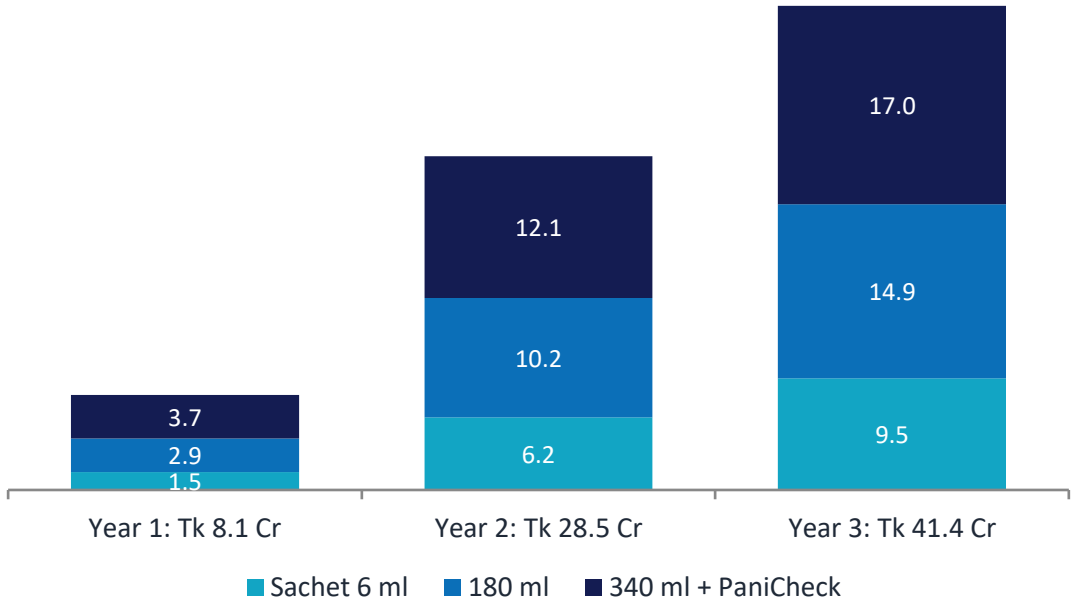
Revenue built bottom-up: outlets × units × net realisation = Tk 41 Cr by Year 3

Where Tk 360 goes: one 340 ml bottle



Gross margin 55% on the 340 ml; blended 47% with the 180 ml (48%) and sachets (29%).

Net revenue by pack (Tk Cr)



Units (million): sachets 9.0 → 36.4 → 56.1 · 180 ml 0.27 → 0.96 → 1.40 · 340 ml 0.17 → 0.57 → 0.80. Year 1 has nine selling months after the pilot and pipeline fill.

Team model v4. Gross-to-net as taught in the Finance masterclass: MRP − VAT − trade margin − SD = net realisation. VAT and SD at 15% on shampoo (VAT & SD Act 2012); invoice treatment to be confirmed with Marico Tax.

P&L: EBITDA-positive in Year 2 and NPV +Tk 3.1 Cr at a 15% hurdle

Tk crore	Year 1	Year 2	Year 3
Gross sales (MRP)	13.8	48.7	70.8
Net revenue	**8.1**	**28.5**	**41.4**
Gross profit	**3.9**	**13.4**	**19.3**
Gross margin	47.9%	47.1%	46.8%
Advertising and promotion	(4.5)	(4.3)	(5.0)
Selling and distribution	(1.6)	(3.3)	(4.3)
EPR (plastic)	(0.1)	(0.3)	(0.4)
EBITDA, standalone	(2.3)	5.6	9.7
Naturale cannibalisation	(0.7)	(2.6)	(3.7)
Incremental EBITDA	**(3.1)**	**3.0**	**5.9**
EBITDA margin	-38.0%	10.6%	14.4%
Depreciation	(0.6)	(0.6)	(0.6)
WPPF and tax	0.0	(0.6)	(1.3)
Net profit	**(3.7)**	**1.8**	**4.1**
Free cash flow	(4.9)	0.9	3.1

NPV @ 15%

+Tk 3.1 Cr

+Tk 0.5 Cr without working-capital recovery

IRR

25.1%

17.0% without recovery; hurdle 15%

Peak funding

Tk 9.8 Cr

Month 25, end of the launch year

Payback

4.8 years

From project start; 3.6 years after first sale

Scenario (Finance masterclass grid)	NPV
Worst: volume −20%, price −5%, COGS +8%, 3-month delay	−Tk 4.4 Cr
Base plan	**+Tk 3.1 Cr**
Best: volume +15%, price +2%, COGS −3%	+Tk 7.6 Cr

The downside is why we pilot first: only Tk 3.5 Cr is at risk before the go/no-go.

Five commercial years after a 12-month R&D and pilot phase; no terminal value; working capital 12% of net revenue. Cannibalisation: 20% of WaterSmart revenue taken from Naturale at 45% contribution. Tax 20%, WPPF 5%. Inputs are planning assumptions (Appendix A).

THE DECISION

Approve a 6-month pilot, and scale nationally only on proof

M1–6 R&D and qualification
Formula, stability, strip shelf-life, BSTI and DGDA

M7–12 Pilot in two cells
~1,500 outlets: a Dhaka hall/mess catchment and one hard-water town

M13–16 National launch
Pipeline fill; first sales in month 16

Year 2 LonaJol gate
Coastal saline variant only after its own tests

GO / NO-GO GATES AT MONTH 12

- **Rate of sale at least 70% of plan**
- **Paid repeat at least 25% in mature cohorts**
- **Blended gross margin at least 45%**
- **Strip complaints under 2%; all safety and regulatory gates passed**

Top risks	Mitigation
Claims not proven	R&D tress and lather tests before any claim; never “stops hair fall”
Strip life in humid storage	Foil-sealed strips, supplier stability data, two suppliers
Cannibalising Naturale	Priced above core Naturale; target users of other brands; measured in the pilot
Cost inflation (COGS +8% cuts NPV by Tk 3.3 Cr)	Dual sourcing; pack and sachet-film optimisation
Regulation: draft cosmetics rules, EPR, sachets	Marico Regulatory sign-off; recyclable mono-material sachet plan

THE ASK

Tk 3.5 Cr

committed before the go/no-go: CAPEX Tk 3.0 Cr + R&D and pilot Tk 0.5 Cr

দোষ কি চুলের, নাকি পানির? আগে পানি চেক করুন।

APPENDIX A

Assumption register: every driver, its basis and who confirms it

Driver	Value	Basis / source	Confirms
MRP 340 ml + PaniCheck	Tk 360	Naturale Hijab Fresh 340 ml Tk 390; Sunsilk 340 ml Tk 350 (Chaldal, 9 Oct 2026)	Brand
MRP 180 ml / 6 ml sachet	Tk 180 / Tk 3	Naturale 160–175 ml Tk 140–150; Naturale sachet Tk 2 per 5.25 ml	Brand
Trade margin	22% bottles, 25% sachets	Retailer + distributor; planning assumption	Sales
VAT / SD	15% / 15%	VAT & SD Act 2012, shampoo; invoice treatment to confirm	Tax
COGS per unit	Tk 95 / 55 / 1.20	Bulk Tk 135/kg plus pack and conversion; strip Tk 3 target (unquoted)	Procurement
Active outlets	37k / 72k / 104k	Channel plan; Sales masterclass universe of 294k outlets	Sales
Units per outlet per month	See Sales slide	Planning assumption; validated by pilot rate of sale	Sales
A&P	Y1 Tk 4.5 Cr; then 15%, 12%, 10%, 10% of NR	Marico FY26 marketing, selling and distribution = 5.2% of revenue	Marketing
Selling and distribution	8% of NR + Tk 1 Cr a year	Incremental SKU on existing Naturale routes	Sales / Finance
Cannibalisation	20% × 45% contribution	Planning assumption; measured in the pilot	Finance / Insights
EPR	1% of NR	EPR Guidelines 2026 (SRO 293); reserve, not a compliance quote	ESG
Working capital	12% of annualised NR	Marico FY26 core working capital = 11.5% of revenue	Finance
CAPEX / pre-launch	Tk 3.0 Cr / Tk 0.5 Cr	Cap mould, strip insertion, line changeover; R&D and pilot	Manufacturing
Tax / WPPF / hurdle	20% / 5% / 15%	Marico FY26 disclosures; hurdle assumed	Finance
Horizon	12 + 60 months	Pre-launch plus five commercial years; no terminal value	Finance

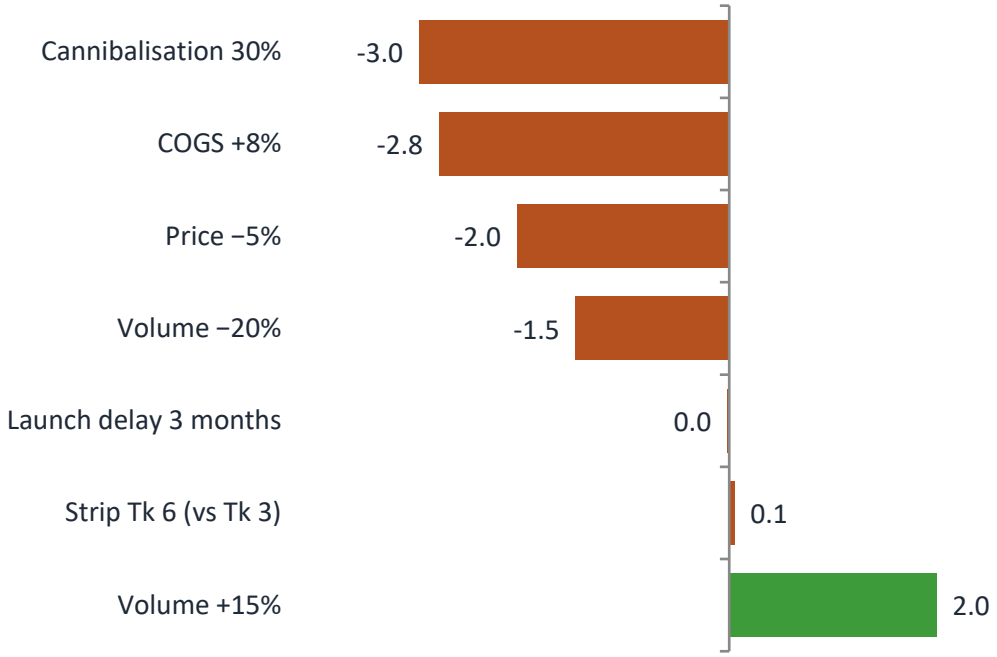
Model: Round2_Submission/finance/finance_v4.cjs. It reuses the team's v3 monthly model unchanged, adds the SD/tax/WPPF correction (reproducing the corrected −3.82/−6.32 Cr), then applies the v4 proposal decisions above.

Evidence base, method and sensitivity

Water: BGS & DPHE (2001) National Hydrochemical Survey: 3,534 wells, 61 districts, sampled Mar 1998–Dec 1999. Team analysis: hardness = 2.497 Ca + 4.118 Mg (USGS); flags Fe >0.3 and hardness >120 mg/L. Historical wells, not today's taps, which is why the pack carries a test.
Recent water: ICCEI 2025 Dhaka tap study (12 locations with values); BWDB 2022 groundwater-quality report (901 monitoring wells, 2019–21).
Consumers: Khanam & Rahman 2022 (n=390); Zakaria et al. 2025 (n=391); team Google Form, 3–9 Oct 2026: 49 consenting adults from 50 submissions, convenience sample. Q20: median maximum Tk 320 (n=47); 15/47 maxima at least Tk 360.
Hair science: Luqman 2018 (n=70: tensile strength 255 vs 234 with hard water); Srinivasan 2013 (n=15: no significant difference). Evidence is mixed, so claims wait for our own R&D tests.
Market and prices: Mordor Intelligence (2026e); 24 online price observations, Chaldal and Arogga, 9 Oct 2026; Marico Bangladesh Annual Report FY2025-26.

R&D TEST PLAN Hair-tress deposit test: at least 30% less Ca/Fe than Naturale core. Lather at least 80% of a soft-water reference at 150 and 300 mg/L. Home-use test n≈100. Stability at 40°C / 75% RH.

NPV without working-capital recovery (base +Tk 0.5 Cr), one change at a time



Biggest levers: cannibalisation and COGS. Both are measured in the pilot before any national spend.

Hair-science references: Luqman et al. 2018 (PMID 30034190); Srinivasan et al. 2013 (PMID 24574692). Sensitivities from finance_v4.cjs.